03/12/2026: Variational Autoencoders

Reference: *Autoencoding Variational Bayes*

CSCI 546: Diffusion Models

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Course Logistics

April 17th-19th 2026

ASA DataFest™
Montana State

Join us for a weekend of fun data analysis and free food!

For more information and registration scan the QR code.

Registration is free!

All undergraduate and first year masters students are welcome.

For more information contact: Samidha Shetty (samidha.shetty@montana.edu)

Lightning Chats

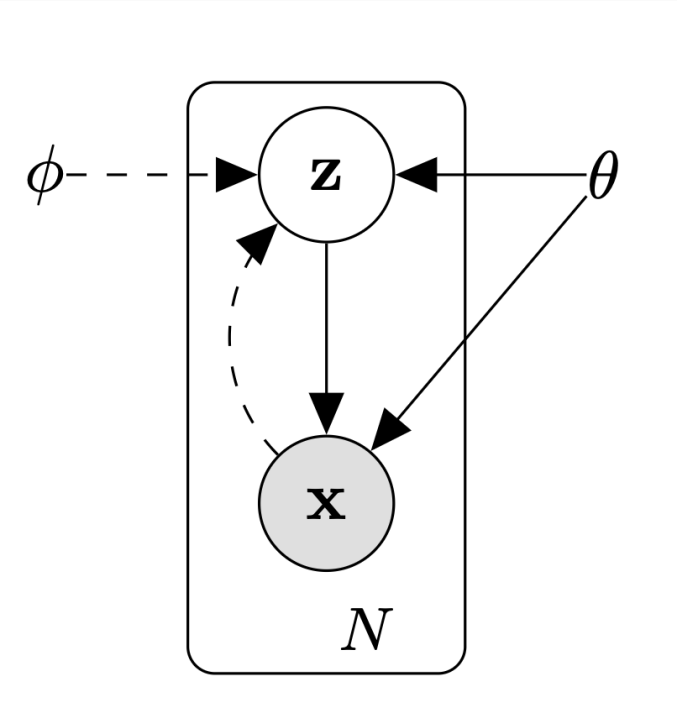
- After Spring Break, any class meeting associated with a reading will have one $\approx$ 10 minute peer **lightning chat**.
- The lightning chats will be given by groups of two. I will do the best I can to assign groups based on your preferences.
- Sometime **today**, please either hand me or email me a list of three readings that are particular interest to you. You can order them if you want. You can name people you'd like to work with.
- If I don't get your preferences today, I will randomly assign you to a partner and paper.

Reading Survey

1. What is a <u>variational</u> autoencoder?

Overview of VAEs

Goal



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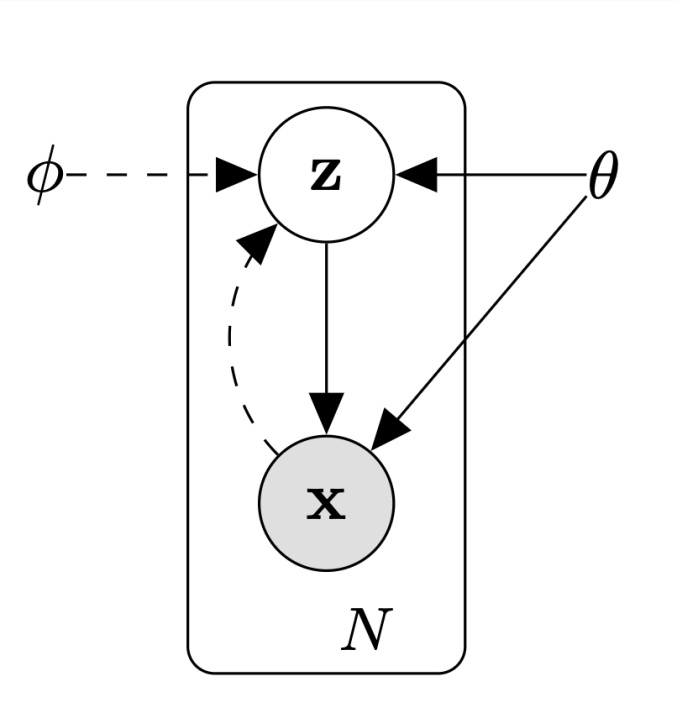

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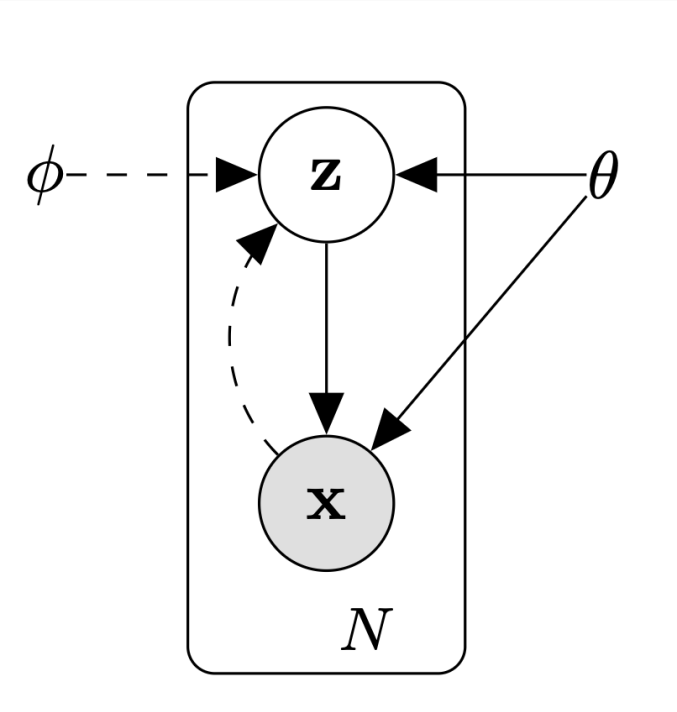

Goal. Learn a probabilistic model giving a low-dimensional latent variable ($\boldsymbol{z}$) representation of the data ($\boldsymbol{x}$) distribution, which allows:

- *Reconstruction*: Can the model reproduce something it has seen?
- *Generation*: Can the model create new examples that look like the dataset?

Comparison to previous approaches

We want the latent "code" to

1. Capture nonlinear structure in the data, going beyond linear dimensionality reduction methods like principal component analysis, and
2. Support probabilistic generative model of the data.

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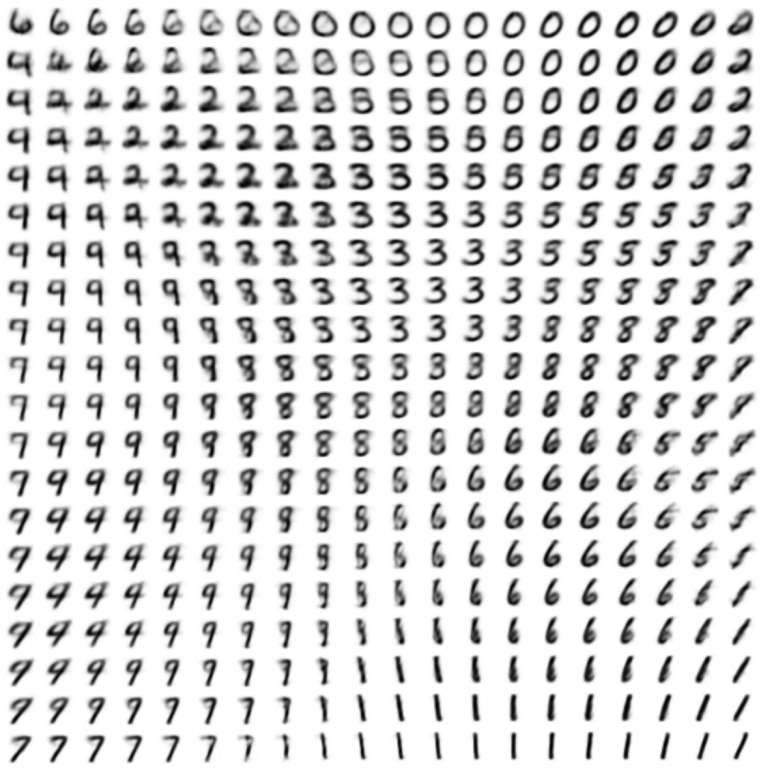

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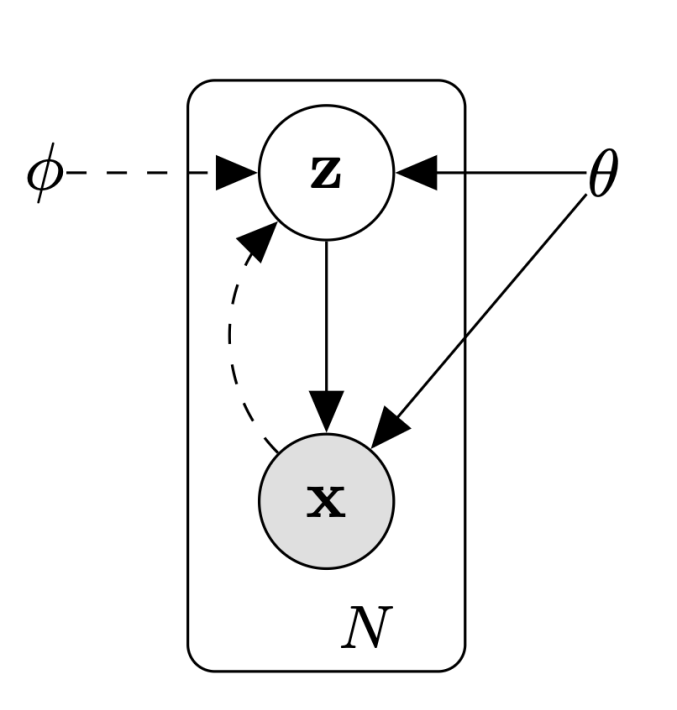

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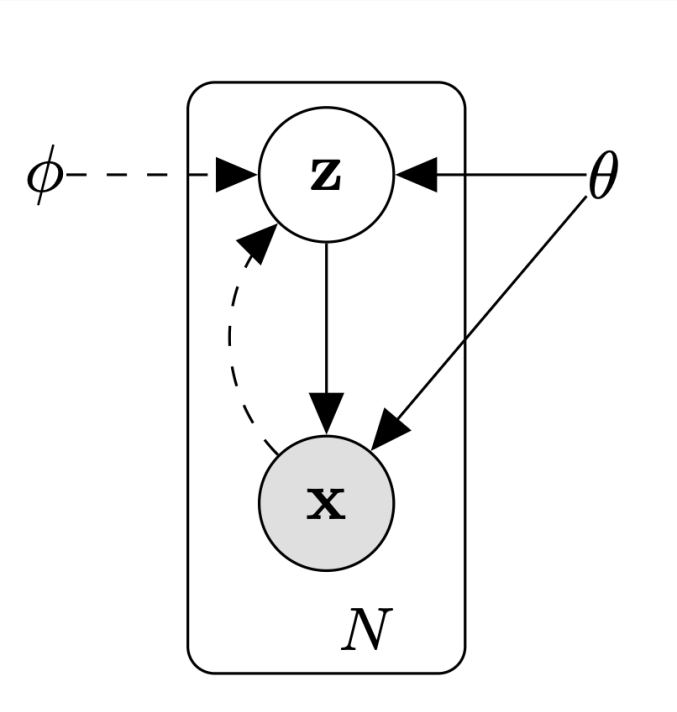


Encoder

$$(\boldsymbol{\mu}_{\boldsymbol{x}}, \log \boldsymbol{\sigma}_{\boldsymbol{x}}) = \texttt{NeuralNetwork}_{\phi}(\boldsymbol{x})$$

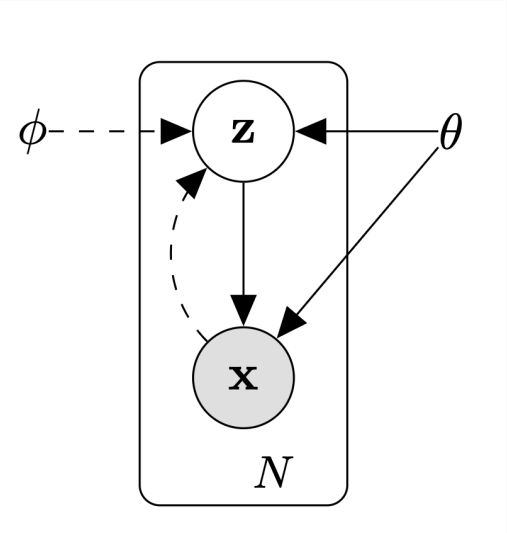
$$q_{\phi}(\boldsymbol{z} \mid \boldsymbol{x}) = \mathcal{N}\left(\boldsymbol{z}; \boldsymbol{\mu}_{\boldsymbol{x}}, \mathrm{diag}(\boldsymbol{\sigma}_{\boldsymbol{x}}^2)\right)$$

Decoder

$$(\boldsymbol{\mu}_{\boldsymbol{z}}, \log \boldsymbol{\sigma}_{\boldsymbol{z}}) = \texttt{NeuralNetwork}_{\theta}(\boldsymbol{z})$$

$$p_{\theta}(\boldsymbol{x} \mid \boldsymbol{z}) = \mathcal{N}\left(\boldsymbol{x}; \boldsymbol{\mu}_{\boldsymbol{z}}, \mathrm{diag}(\boldsymbol{\sigma}_{\boldsymbol{z}}^2)\right)$$

Bayesian Perspective



$$p(\boldsymbol{z}) = \mathcal{N}\left(\boldsymbol{z}; 0, \boldsymbol{I}\right) \qquad \textbf{Prior}$$

$$p_\theta(\boldsymbol{x} \mid \boldsymbol{z}) = \mathcal{N}\left(\boldsymbol{x}; \boldsymbol{\mu}_{\boldsymbol{z}}, \mathrm{diag}(\boldsymbol{\sigma}_{\boldsymbol{z}}^2)\right) \qquad \textbf{Likelihood}$$

$$q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) = \mathcal{N}\left(\boldsymbol{z}; \boldsymbol{\mu}_{\boldsymbol{x}}, \mathrm{diag}(\boldsymbol{\sigma}_{\boldsymbol{x}}^2)\right) \qquad \textbf{(Variational) Posterior}$$

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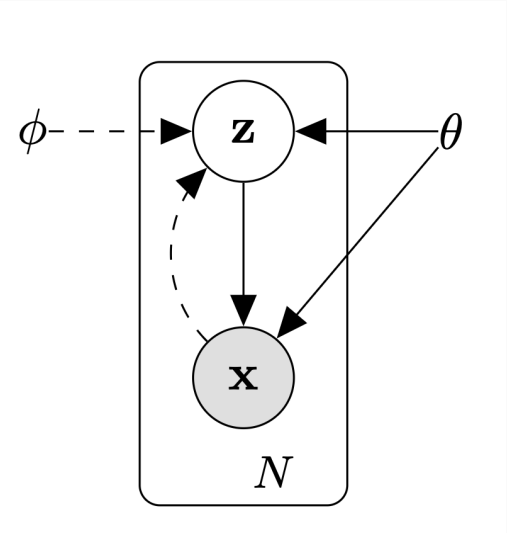

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Remark. The prior here is serving a different purpose than classical Bayesian models.

- In **classical Bayes**, the prior expresses knowledge about the world.
- In a **VAE**, the prior expresses how we want the latent space to be organized.

Gaussian latent variable models

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Gaussian latent variable models

We can interpret a VAE as a `nonlinear` Gaussian latent variable model.

Classical `linear` Gaussian latent variable model

In models like probabilistic Principal Components Analysis (PCA):

$$\boldsymbol{z} \sim \mathcal{N}(0, \boldsymbol{I})$$
$$\boldsymbol{x} \mid \boldsymbol{z} \sim \mathcal{N}(\boldsymbol{W}\boldsymbol{z} + \boldsymbol{b}, ; \sigma^2 \boldsymbol{I})$$

Key properties:

- Latent variable ($\boldsymbol{z}$) is Gaussian
- Observation ($\boldsymbol{x}$) is Gaussian with a mean that is a linear function of $\boldsymbol{z}$:

$$\boldsymbol{\mu}_{\boldsymbol{z}} = \boldsymbol{W}\boldsymbol{z} + \boldsymbol{b}$$

Gaussian latent variable models

We can interpret a VAE as a `nonlinear` Gaussian latent variable model.

VAE as a `nonlinear` Gaussian model.

In a VAE:

$$\boldsymbol{z} \sim \mathcal{N}(0, \boldsymbol{I})$$
$$\boldsymbol{x} \mid \boldsymbol{z} \sim \mathcal{N}\big(f_\theta(\boldsymbol{z}), \mathrm{diag}(g_\theta(\boldsymbol{z}))\big)$$

where now f_θ and g_θ are neural networks.

So the mean and variance of the Gaussian observation model are non-linear functions of the latent variable.

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Poll. So what?

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Solution.

- Linear Gaussian models imply the data lies near a **linear subspace**.
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Solution.

- Linear Gaussian models imply the data lies near a **linear subspace**.
- A VAE instead learns a **nonlinear manifold** in data space.

Overall. A VAE is a nonlinear Gaussian latent variable model, where the mean and variance of the observation distribution are parameterized by neural networks.

Inference

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Review: The CAVI Algorithm

Mean Field Coordinate Ascent Variational Inference (MF-CAVI)

Mean field variational families

A variational family $\mathscr{Q}$ is mean field if it factorizes

$$q(u_1, ..., u_K) = \prod_{k=1}^{K} q_k(u_k) \tag{1}$$

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Remark. Since the ELBO maps probability densities to a real number, it is a functional, so such optimization can be handled with variational calculus.

Update equations for MF-CAVI

Given the mean field assumption (1), the optimal updates to $\{q_k\}_k$ for coordinate ascent on the ELBO satisfy

$$q_k(u_k) \propto \exp\left\{\mathbb{E}_{q_{-k}}\left[\log p(u_k \mid \boldsymbol{u}_{-k}, \boldsymbol{x})\right]\right\} \tag{2}$$

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Remark. We can also write the update as

$$q_k(u_k) \propto \exp\left\{\mathbb{E}_{q_{-k}}\left[\log p(\boldsymbol{u}, \boldsymbol{x})\right]\right\} \tag{3}$$

which is often referred to as a geometric mean.

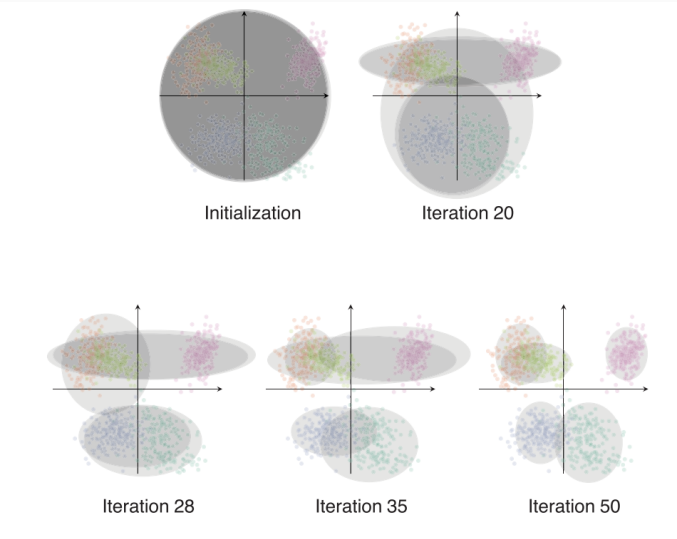
Example: Bayesian Gaussian Mixture Model

To see the mean field CAVI algorithm (3) in a concrete context, consider a version of the Bayesian Gaussian Mixture Model.

$$\begin{aligned} \mu_k &\sim \text{Normal}(0, \sigma^2) && k = 1, \ldots, K \\ \boldsymbol{c}_i &\sim \text{Categorical}(\pi_1, \ldots, \pi_K) && i = 1, \ldots, n \\ x_i \mid \boldsymbol{c}_i, \boldsymbol{\mu} &\sim \text{Normal}(\boldsymbol{c}_i^\top \boldsymbol{\mu}, 1) && i = 1, \ldots, n \end{aligned}$$

(This simplified version assumes that each mixture component has unit variance.)

Poll. Who can explain the variables in this model? What do we want to learn?



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The joint density, by chain rule, is

$$p(\boldsymbol{x}, \boldsymbol{c}, \boldsymbol{\mu}) = p(\boldsymbol{\mu}) \prod_{i=1}^{n} p(\boldsymbol{c}_i) p(x_i \mid \boldsymbol{c}_i, \boldsymbol{\mu})$$

And a mean-field variational family is given by

$$q(\boldsymbol{c}, \boldsymbol{\mu}) = \prod_{k=1}^{K} q(\mu_k) \prod_{i=1}^{n} q(\boldsymbol{c}_i)$$

Example: Bayesian Gaussian Mixture Model

We apply (3) to determine the coordinate updates for q_{c_i}, the variational factors governing cluster assignments. We take the log of the joint and discard terms that do not depend upon c_i to obtain

$$
\begin{aligned}
q(c_{ik}) &\propto \exp\left\{\mathbb{E}_{q_{\mu_k}}\left[\log p(c_i = k) + \log p(x_i \mid c_i = k, \mu)\right]\right\} \\
&\propto \exp\left\{\mathbb{E}_{q_{\mu_k}}\left[\log \pi_k + x_i \mu_k - \frac{1}{2}\mu_k^2\right]\right\} \\
&\propto \pi_k \exp\left\{x_i \mathbb{E}_{q_{\mu_k}}[\mu_k] - \frac{1}{2}\mathbb{E}_{q_{\mu_k}}[\mu_k^2]\right\}
\end{aligned}
$$

The coordinate updates for q_{μ_k} are derived similarly. They reveal that q_{μ_k} are Gaussian, and hence the above expectations are easy to compute.

Bayesian Gaussian Mixture Model: Updates to mixture component means

Using the same strategy as when updating cluster assignments c_i, we obtain

$$
\begin{aligned}
q(\mu_k) &\propto \exp\left\{\mathbb{E}_{-q_{\mu_k}}\left[\log p(\mu_k) + \sum_{i=1}^{n} \log p(x_i \mid c_i = k, \mu)\right]\right\} \\
&\propto \exp\left\{-\frac{1}{2\sigma^2}\mu_k^2 + \sum_{i=1}^{n} \mathbb{E}_{q_c}\left[1_{c_i=k}\left(x_i\mu_k - \frac{1}{2}\mu_k^2\right)\right]\right\} \\
&\propto \exp\left\{-\frac{1}{2}\left(\frac{1}{\sigma^2} + \sum_{i=1}^{n} q(c_{ik})\right)\mu_k^2 + \left(\sum_{i=1}^{n} q(c_{ik})x_i\right)\mu_k\right\}
\end{aligned}
$$

which is an exponential family distribution with sufficient statistics (μ_k, μ_k^2), and hence Gaussian.

Bayesian Gaussian Mixture Model: CAVI Algorithm

Algorithm 2: CAVI for a Gaussian mixture model

Input: Data $x_{1:n}$, number of components K, prior variance of component means σ^2

Output: Variational densities $q(\mu_k; m_k, s_k^2)$ (Gaussian) and $q(z_i; \varphi_i)$ (K-categorical)

Initialize: Variational parameters $\mathbf{m} = m_{1:K}$, $\mathbf{s}^2 = s_{1:K}^2$, and $\boldsymbol{\varphi} = \varphi_{1:n}$

while *the ELBO has not converged* **do**

  for $i \in \{1, \ldots, n\}$ **do**

    Set $\varphi_{ik} \propto \exp\{\mathbb{E}[\mu_k; m_k, s_k^2]x_i - \mathbb{E}[\mu_k^2; m_k, s_k^2]/2\}$

  end

  for $k \in \{1, \ldots, K\}$ **do**

    Set $m_k \longleftarrow \dfrac{\sum_i \varphi_{ik} x_i}{1/\sigma^2 + \sum_i \varphi_{ik}}$

    Set $s_k^2 \longleftarrow \dfrac{1}{1/\sigma^2 + \sum_i \varphi_{ik}}$

  end

  Compute $\textsc{elbo}(\mathbf{m}, \mathbf{s}^2, \boldsymbol{\varphi})$

end

return $q(\mathbf{m}, \mathbf{s}^2, \boldsymbol{\varphi})$

Inference

A more flexible algorithm: Stochastic Gradient Variational Bayes

The problem with CAVI

Problem. The CAVI updates are:

$$q_k(u_k) \propto \exp\left\{\mathbb{E}_{q_{-k}}\left[\log p(u_k \mid \boldsymbol{u}_{-k}, \boldsymbol{x})\right]\right\}$$

What if we can't compute these expectations analytically?

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Big-Picture Idea. Let's try to use (stochastic) gradient descent.

Older idea

Recall. The objective we're trying to optimize is

$$\texttt{ELBO}(q_\phi) = \underbrace{\mathbb{E}_{q_\phi}\left[\log p(\boldsymbol{x}, \boldsymbol{z})\right]}_{\text{energy}} + \underbrace{-\mathbb{E}_{q_\phi}\left[\log q_\phi(\boldsymbol{z})\right]}_{\text{entropy}}$$

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Poll. We'd like to take the gradient $\nabla_\phi \texttt{ELBO}(q_\phi)$. Does anybody see a problem?

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An older idea. Using the log-derivative trick, we can write

$$\begin{aligned}\nabla_\phi \mathbb{E}_{q_\phi(\boldsymbol{z})}[f(\boldsymbol{z})] &= \mathbb{E}_{q_\phi(\boldsymbol{z})}\left[f(\boldsymbol{z})\nabla_\phi \log q_\phi(\boldsymbol{z})\right] \\ &\approx \frac{1}{S}\sum_{s=1}^{S} f(\boldsymbol{z}^{(s)})\nabla_\phi \log q_\phi(\boldsymbol{z}^{(s)}), \qquad \boldsymbol{z}^{(s)} \sim q_\phi\end{aligned}$$

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Problem. This gradient estimator has such high variance that it's unusual.

Their idea: Stochastic Gradient Variational Bayes

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Their idea. They use the "reparametrization trick". They reparameterize the random variable $\widetilde{\boldsymbol{z}} \sim q_\phi(\boldsymbol{z} \mid \boldsymbol{x})$ using a differentiable transformation $g_\phi(\boldsymbol{\epsilon}, \boldsymbol{x})$ of an (auxiliary) noise variable $\boldsymbol{\epsilon}$:

$$\widetilde{\boldsymbol{z}} = g_\phi(\boldsymbol{\epsilon}, \boldsymbol{x}) \qquad \text{with} \qquad \boldsymbol{\epsilon} \sim p(\epsilon)$$

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Who cares? Then they can write

$$\begin{aligned}\mathbb{E}_{q_\phi(\boldsymbol{z}|\boldsymbol{x}_i)}[f(\boldsymbol{z})] &= \mathbb{E}_{p(\boldsymbol{\epsilon})}\left[f\big(g_\phi(\boldsymbol{\epsilon}, \boldsymbol{x}_i)\big)\right] \\ &\approx \frac{1}{S}\sum_{s=1}^{S} f\big(g_\phi(\boldsymbol{\epsilon}^{(s)}, \boldsymbol{x}_i)\big), \qquad \epsilon^{(s)} \sim p(\boldsymbol{\epsilon})\end{aligned}$$

Good, Bad, Ugly, Beautiful

Good, Bad, Ugly, Beautiful

Guidelines

1. *Good, Bad*: Technical content
2. *Ugly, Beautiful*: Presentational styles

Format

1. Small group discussion
2. Class discussion